

Isaac Mubiru

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Kampala, Uganda (EAT, UTC+3) — Fully Remote

PROFESSIONAL SUMMARY

Senior backend engineer with 6+ years of production Node.js — specialized in **real-time systems, AI-assisted features, and zero-downtime reliability**. Shipped a **real-time market-data pipeline at Dr Wealth** (Morningstar → Firestore → live PWA, 2M+ records), **DeFi backends at Mind2matter** using Web3 + Node.js, and a **healthcare WhatsApp chatbot integrating OpenAI + Google Vision**. At MTailor, led the end-to-end zero-downtime migration of 20+ production apps, saving \$5K/month. Comfortable with Node.js, Python, React, Pub/Sub messaging, Docker/Kubernetes, and the Firebase/Supabase serverless-Postgres paradigm.

CORE COMPETENCIES

Real-time systems

Node.js + TypeScript

AI / LLM integration

Pub/Sub architectures

Web3 / DeFi backends

Zero-downtime migrations

Postgres / Supabase-ready

Docker + Kubernetes

Performance & observability

EXPERIENCE

MTailor

Sep 2022 – Present · US Remote

Software Engineer — reporting to CTO

- Led zero-downtime end-to-end migration of 20+ production apps from Parse / MongoDB to Firebase / GCP; kept live traffic flowing throughout.
- Implemented **real-time two-way sync between MongoDB and Firestore** using Node.js + Google Pub/Sub — same class of reliability pattern used in trading-platform backends.
- Saved the company **\$5,000/month** by migrating all services off AWS.
- Shipped 3D product visualization (ffmpeg video overlay + React UI) that lifted buyer conversion.
- Implemented Express Shipping — generated an additional \$40 revenue per order.

Node.js · Firebase · GCP Pub/Sub · AWS S3/EC2 · Python · React

Dr Wealth

Aug 2021 – Jan 2022 · Singapore Remote

Software Engineer — real-time market-data systems

- Led optimization of a PWA delivering **real-time stock-market prices** from a Firebase backend — directly analogous to trading-platform market-data feeds.
- Built ad-hoc jobs to query and reconcile over **2 million Firestore records**, keeping customer prices current from the Morningstar API.
- Extended backend APIs using Firebase Cloud Functions + Express.js on Heroku.
- Shipped responsive React / Tailwind UI pages.

Node.js · Firebase · React · Morningstar API · Tailwind

Mind2matter

Jan 2022 – Jul 2022 · US Remote

Software Engineer (Full-stack) — reporting to CTO

- Built **backends for DeFi applications using Web3 + Node.js** — direct experience shipping into the digital-asset domain.
- Delivered React UIs from Figma designs under tight agency deadlines.

Node.js · Web3 · Firebase · React · Tailwind

CodeBits

Jan 2020 – Jul 2021 · Uganda Remote

Software Engineer — led team of 4 developers

- Architected a **microservices backend** for FIDA Uganda's case-management app using **Apache Kafka, Docker, and Kubernetes**.
- Built a Node.js backend serving mobile + web + USSD clients over gRPC.
- Shipped a cross-platform React Native + Expo app (iOS / Android) with Firebase push notifications.

Node.js · Apache Kafka · Docker · Kubernetes · gRPC · React Native · Firebase

MTailor (Contractor)

Mar 2022 – Jul 2022 · US Remote

Backend Contractor (hired via Upwork)

- Built migration scripts reading from MongoDB and writing to Firestore with complex transformation logic.
- Wrote Node.js and Python Firebase-SDK documentation for the engineering team; trained the Ops team on the new dashboard.

Node.js · Python · Firebase · AWS · GCP

SELECTED PROJECTS

Healthcare WhatsApp Chatbot — OpenAI + Google Vision

Integrated OpenAI and Google Vision with the WhatsApp Business API to let healthcare providers OCR prescriptions over WhatsApp. Shipped the full prompt / guardrail / retry loop — directly relevant to "AI-assisted features" in a production system.

OpenAI API · Google Vision · WhatsApp Business API · Node.js

Real-time Market-Data Pipeline (Dr Wealth)

Kept a PWA's stock prices live from the Morningstar API by reconciling 2M+ Firestore records and surfacing them through Firebase Cloud Functions. Same architectural class as a trading platform's market-data layer.

Node.js · Firestore · Morningstar API · Firebase Cloud Functions

DeFi Backends (Mind2matter)

Production backends for decentralized-finance applications using Web3 and Node.js, shipped under agency timelines for a US-based client.

Node.js · Web3 · Firebase · React

SKILLS

Proficient: Node.js · JavaScript · TypeScript · Firebase · GCP · Python · React · MongoDB · SQL · Git · Linux · HTML / CSS

Intermediate: AWS (S3 / EC2 / EBS) · PostgreSQL / Supabase · Go · Rust · React Native · Redis

Production tooling: Apache Kafka · Google Pub/Sub · Docker · Kubernetes · gRPC · ffmpeg

AI / Fintech domain: OpenAI API · Google Vision · WhatsApp Business API · Web3 / DeFi backends · Real-time market-data pipelines

EDUCATION

B.Sc. Computer Engineering — Makerere University, Kampala

Aug 2018 – Sep 2022

LANGUAGES

English · Kiswahili